MILITARY SPECIFICATION SHEET

ELECTRON TUBE, RECEIVING

TYPE 5687WB

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the electron tube described herein shall consist of this document and the latest issue of Specification MIL-E-1.

DESCRIPTION: Double triode, miniature, low Mu

| | | |
|---|---|---|
| Outline | --- | 6-2 (EIA) |
| Base | --- | E9-1 |
| Envelope | --- | T6-1/2 |
| Cathode | --- | Coated unipotential |

Base connections:

| Pin No. | --- | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 |
|---|---|---|---|---|---|---|---|---|---|---|
| Element | --- | 2a | 2g | 2k | h | h | 1k | 1g | hct | 1a |

ABSOLUTE-MAXIMUM RATINGS:

Normal

| Parameter: | Ef | Eb | Ec | Ehk | Rg/g | Ic/g | Ik/k | Pp/p | TE | Alt |
|---|---|---|---|---|---|---|---|---|---|---|
| Unit: | V | Vdc | Vdc | v | Meg | mAdc | mAdc | W | °C | ft |
| Maximum: | 6.6 | 330 | 0, -200 | 135 | 0.1 | 6.0 | 65 | 3.75 | 225 | (see note 3) |
| | 13.2 | --- | --- | --- | (see note 1) | --- | --- (see note 2) | | --- | --- |
| Minimum: | 6.0 | --- | --- | --- | --- | --- | --- | --- | --- | --- |
| | 12.0 | --- | --- | --- | --- | --- | --- | --- | --- | --- |
| TEST CONDITION (1): | 12.6 | 120 | -2 | 0 | --- | --- | --- | --- | --- | --- |

Pulse service

| Parameter: | Ef | Eb | Ec | eb | egk | ehk | Pg/g | Pp/p | ik/k | Du | tp | Prr |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Unit: | V | Vdc | Vdc | v | v | v | W | W | ma | % | µs | |
| Maximum: | 6.6 | 330 | --- | 660 | 50, -100 | 100 | 0.1 | 4.0 | See figure 1 | | | |
| | 13.2 | --- | --- | --- | --- | --- | --- | --- | | | | |
| Minimum: | 6.0 | --- | --- | --- | --- | --- | --- | --- | | | | |
| | 12.0 | --- | --- | --- | --- | --- | --- | --- | | | | |
| TEST CONDITIONS (2): | 12.6 | 300 | -40 | --- | 40 | --- | --- | --- | --- | 1.0 | 10 | 1,000 |

GENERAL:

Qualification - Required

Reliable tube



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| METHOD | REQUIREMENT OR TEST | CONDITIONS | AQL PERCENT DEFECTIVE | INSP LEVEL OR CODE | SYMBOL | LIMITS MIN | LIMITS MAX | UNIT |
|---|---|---|---|---|---|---|---|---|
| | Quality conformance inspection, part 1 | | | | | | | |
| 1256 | Electrode current (1) (anode) | Test condition (1) (see notes 4 and 5) | 0.4 | II | Ib | 27 | 45 | mAdc |
| 1266 | Total grid current | Test condition (1) (see notes 4 and 5) | 0.4 | II | Ic | 0 | -1.5 | µAdc |
| 1301 | Heater current | | 0.4 | II | If | 410 | 470 | mA |
| 1306 | Transconductance (1) | Test condition (1) (see note 4) | 0.4 | II | Sm | 8500 | 14500 | µmhos |
| 1336 | Heater-cathode leakage | See note 4 | 0.4 | II | Ihk | --- | 30 | µAdc |
| 1201 | Short and discontinuity detection | | 0.4 | II | --- | --- | --- | --- |
| | Quality conformance inspection, part 2 | | | | | | | |
| 1211 | Insulation of electrodes | E(g to all) = -300 V<br>E(a to all) = -500 V<br>(see note 4) | 2.5 | I | --- | --- | --- | --- |
| 1031 | Low-frequency vibration | Test condition (1); F = 40 Hz; 10 G; Rp = 2,000 ohms (see note 4) | 6.5 | Code H | Ep | --- | 100 | mVac |
| 2201 | Noise and microphonics | Ebb = 300 Vdc; Ec = 0; Esig = 70 mVac; Rk = 680 ohms; Rg = 1.0 Meg; Rp = 2,000 ohms (see note 6) | 2.5 | I | --- | --- | --- | --- |
| 1256 | Electrode current (2) (anode) | Test condition (1); Eb = 300 Vdc; Ec = -20 Vdc (see note 4) | 2.5 | I | Ib | --- | 5.0 | mAdc |
| 1256 | Electrode current (3) (anode) | Test condition (1); Eb = 300 Vdc; Ec = -25 Vdc (see note 4) | 2.5 | I | Ib | --- | 1.0 | mAdc |



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| METHOD | REQUIREMENT OR TEST | CONDITIONS | ACC PERCENT DEFECTIVE | INSP LEVEL OR CODE | SYMBOL | Limits Min | Limits Max | Unit |
|---|---|---|---|---|---|---|---|---|
| | Quality conformance inspection, part 2 -Continued | | | | | | | |
| 1266 | Grid emission | Ef = 14.0 V; Eb = 120 Vdc; Ec = -25 Vdc (see notes 4 and 8) | 2.5 | I | Is | 0 | -5.0 | μAdc |
| 1296 | Pulse current (1) (cathode) | Test condition (2); Rk/k = 1.0 ohm (see notes 4 and 9) | 2.5 | Code H | ik | 900 | --- | ma |
| 1296 | Pulse current (2) (cathode) | Test condition (2); Ef = 12.0 V; Rk/k = 1.0 ohm (see notes 4 and 9) | 6.5 | Code H | ik | 800 | --- | ma |
| 1306 | Transconductance (2) | Test condition (1); Ef = 11.4 V (see note 4) | 2.5 | I | ΔSm Ef | --- | 15 | % |
| 1316 | Amplification factor | Test condition (1) (see note 4) | 2.5 | I | Mu | 16 | 21 | --- |
| 1331 | Direct-interelectrode capacitance | No shield (see note 4) | 6.5 | Code E | Cgp | 2.8 | 5.2 | pF |
| | | No shield (see note 4) | | | Cin | 2.8 | 5.2 | pF |
| | | No shield (triode No. 1) | | | Cout | 0.42 | 0.78 | pF |
| | | No shield (triode No. 2) | | | Cout | 0.34 | 0.66 | pF |
| | | No shield (see note 4) | | | Chk | --- | 9.7 | pF |
| 1231 | Emission current (anode) | Epp = 195 Vac; Ec = 0; Rk/Ib = 10.5 mAdc (see note 10) | 2.5 | I | Is | --- | 25 | μAdc |
| 1121 | Base strain | See note 11 | --- | --- | --- | --- | --- | --- |
| 2126 | Glass strain | | 2.5 | I | --- | --- | --- | --- |
| 1041 | Shock | Test condition (1); 450 G; Ehk = 100 Vdc (see note 12) | 6.5 | See note 13 | --- | --- | --- | --- |
| 1031 | Vibration fatigue | 2.5 G; F = 25 (min), 60 (max); fixed frequency | 6.5 | See note 13 | --- | --- | --- | --- |
| --- | Post-shock and vibration-fatigue test end points | | | | | | | |
| 1031 | Low-frequency vibration | | --- | --- | Ep | --- | 150 | mVac |
| 1336 | Heater-cathode leakage | | --- | --- | Ihk | --- | 50 | μAdc |
| 1306 | Transconductance (1) | | --- | --- | Sm | 6500 | --- | μmhos |
| 1266 | Total grid current | | --- | --- | Ic | --- | -3.0 | μAdc |
| 1105 | Permanence of marking | | --- | --- | --- | --- | --- | --- |



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| METHOD OR PARAGRAPH | REQUIREMENT OR TEST | CONDITIONS | SYMBOL | LIMITS | | UNIT |
|---|---|---|---|---|---|---|
| | | | | MIN | MAX | |
| | Quality conformance inspection, part 3 | | | | | |
| 1506 | Heater-cycling life | Ef = 7.5 V; 1 min "on", 4 min "off"; Ehk = 135 Vdc; Eb = Ec = 0 (see note 7) | --- | --- | --- | --- |
| --- | Heater-cycling life-test end point | | | | | |
| 1336 | Heater-cathode leakage | | Ihk | --- | 30 | µAdc |
| 1516 | Stability life | Test condition (1); Ec = 0; Rk/k = 68 ohms; Ehk = 135 Vdc; Rg/g = 1.0 Meg; TA = room | --- | --- | --- | --- |
| --- | Stability life-test end point | | | | | |
| 1306 | Change in transconductance (1) of individual tubes | | ΔSm t | --- | 10 | % |
| 1501 | Intermittent life | Stability life-test conditions; TE = 225°C (min) (see notes 14 and 16) | --- | --- | --- | --- |
| --- | Intermittent life-test end points (1,000 hours) | | | | | |
| --- | Inoperatives | | --- | --- | --- | --- |
| 1266 | Total grid current | | Ic | 0 | -2.5 | µAdc |
| 1301 | Heater current | | If | 400 | 480 | mA |
| 1306 | Change in transconductance (1) of individual tubes | | ΔSm t | --- | 25 | % |
| 1306 | Transconductance (2) | | ΔSm Ef | --- | 30 | % |
| 1336 | Heater-cathode leakage | | Ihk | --- | 50 | µAdc |
| 1211 | Insulation of electrodes E (g to all) = -300 V E (a to all) = -500 V | | R | 25 | --- | Meg |
| 1501 | Intermittent life (pulse) | Test condition (2); Rb/p = 50 ohms (see note 15) | --- | --- | --- | --- |
| --- | Intermittent pulse life-test end points | | | | | |
| | Inoperative | | --- | --- | --- | --- |
| 1296 | Pulse cathode current (1) | | Ik | 750 | --- | ma |

NOTES:

1. This value is for operation under fixed-bias conditions. With cathode bias, Rg/g may be 1.0 megohm maximum.
2. Pp/p on one section may be as great as 4.2 watts maximum providing that the total for both sections does not exceed 7.5 watts.
3. See "Reduced pressure (altitude) rating", and altitude, maximum peak voltage.



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NOTES: -Continued

4. Test each section separately.

5. This test shall be performed at the conclusion of the holding period.

6. Tie 1k to 2k; 1g to 2g; and 1a to 2a.

7. Operate heaters in parallel.

8. Prior to this test, tubes shall be preheated a minimum of 5 minutes with all sections operating at the conditions specified below. Test at specified conditions within 3 second after preheating. The 3-minute test shall not be permitted. Grid emission shall be the last test performed on the sample selected for the grid-emission test.

| Ef | Eb | Ec1 | Rk/k | Rg/g |
|---|---|---|---|---|
| V | Vdc | Vdc | Ohms | Meg |
| 14.0 | 120 | -2 | 0 | 1.0 |

9. The positive portion of the grid pulse shall be a square wave meeting pulse shape requirement of method 1296 and, in addition, the maximum amplitude shall occur within the first 20 percent of tp. The pulse shall be applied to the grid by means of a driving circuit which produces the specified peak pulse voltage directly at the grid terminal with respect cathode. Grid resistance, not exceeding 50 ohms may be inserted to prevent oscillation, provided readjustment of grid drive is made to maintain the specified pulse amplitude directly at the grid terminal. Peak currents shall be measured by means of a high impedance oscilloscope, or equivalent device, connected across a cathode resistor of 1.0 $\pm$0.01 ohm. The specified limit refers to the maximum of pulse amplitude.

10. Test each unit separately with test voltages applied to opposite section. Tie grids to negative end of individual Rk through individual 100-ohm resistors. Adjust individual Rk for specified forward anode current $\pm$5 percent as measured on the forward half cycles of anode voltage. After a minimum of 5 minutes operation as above, measure reverse anode current on the reverse half cycles of anode voltage. See figure 2.

11. Acceptance sampling procedure shall be in accordance with "Base-strain test, miniature, sampling (method 1121)", except that data covered in "Acceptance and rejection criteria" shall be modified as follows:

(a) Accepted if not more than one defective for class "A", "B", or "C" defects, respectively (see method 1121), or if not more than a total of two defectives are found in the sample.

(b) Rejected if two or more defectives for class "A", "B", or "C" defects, respectively, or if a total of three or more defectives are found in the sample.

12. A grid resistor of 0.1 megohm shall be added.

13. This test shall be conducted on the initial lot and thereafter on a lot approximately every 12 months. When one lot has passed, the 12-month rule shall apply. In the event of lot failure, the lot shall be rejected and the succeeding lots shall be subjected to this test until a lot passes. MIL-STD-105, sample size code letter E, shall apply.

14. Envelope temperature (TE) requirements, when measured in accordance with the temperature by conduction-band measurement (method 1226), will be satisfied if a tube having bogey Ib ($\pm$5 percent) under normal test conditions, is determined to operate at or above minimum specified temperature at any position in the life-test rack.



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NOTES: -Continued

15. The positive portion of the grid pulse shall be a rectangular wave meeting pulse shape requirement of method 1296. The pulse shall be applied to the grid by means of a driving circuit which produces the specified peak pulse voltage directly at the grid terminal with respect to the cathode. Grid resistance not exceeding 50 ohms may be inserted to prevent oscillation, provided readjustment of grid drive is made to maintain the specified pulse amplitude directly at the grid terminal. The pulse width, tp, shall be 10 $\pm$2 µs, and the duty factor, 0.9 percent to 1.1 percent. Self-excited life-test circuitry is permissible, provided any additional anode voltage drops during the time of the pulse are compensated for by increasing Ebb. No fixed Ecl need be applied under self-excited conditions.

16. The life-test sample shall consist of 20 tubes per lot and not more than 1 tube failure shall be permitted. In the event of rejection of the first sample, due to failure of more than 1 tube, a second sample of 40 tubes shall be selected from the lot. Acceptance shall then be based on the combined first and second samples. The total tube failures from the combined first and second samples shall not exceed three.

17. Revision letters are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

Custodians:
Army - ER
Navy - EC
Air Force - 85

Review activities:
Army - AR, MI
Air Force - 99
DLA - ES

User activities:
Army - ME, AV
Navy - AS, OS, MC
Air Force - 11

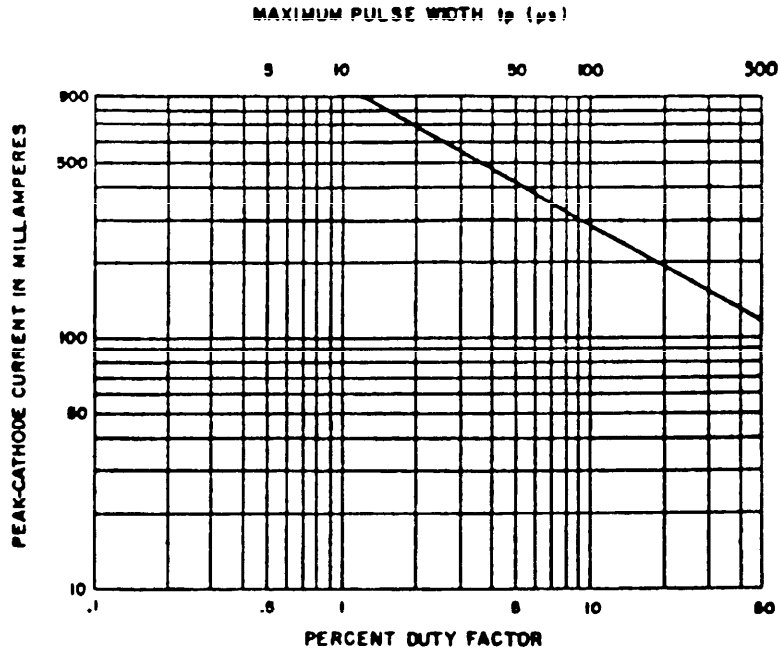
Preparing activity:
Navy - EC

Agent:
DLA - ES

(Project 5960-3224)



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(H) The area below and to the left of the line is the area of permissible operation. Nine hundred milliamperes is the maximum peak current at any duty factor less than 1 percent. For any tube operating under the worst probable conditions, the point indicating peak cathode current and percent duty factor and the point indicating peak-cathode current and pulse width should both be in the area of permissible operation. Duty factor is defined as the ratio of the average current to the maximum peak current occurring in any 1,000-microsecond period.

FIGURE 1. Pulse rating chart.

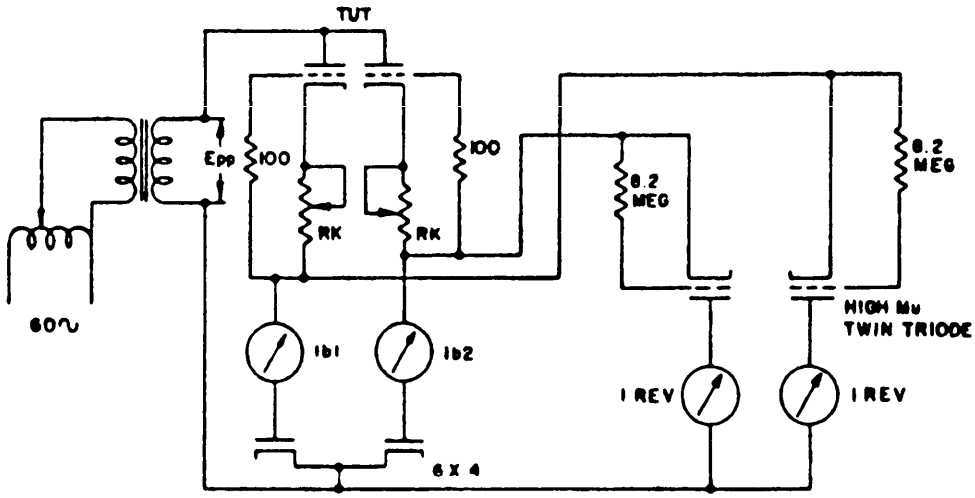



FIGURE 2. Anode emission test circuit.

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